

Quantum mechanics for nuclei

W. Ryssens



Many-body quantum mechanics

Describing a single nucleon

Four measurable quantities = four Hermitian operators

- position $\hat{r}$
- momentum $\hat{p}$
- spin $\hat{s} = \frac{1}{2}\vec{\sigma}_{\text{spin}}$
- isospin $\hat{\tau} = \frac{1}{2}\vec{\sigma}_{\text{isospin}}$

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Their (non-trivial) commutation relations; $\mu, \nu, \kappa = x, y, z$

$$[\hat{r}_\mu, \hat{p}_\nu] = i\hbar\delta_{\mu\nu} . \quad [\hat{s}_\mu, \hat{s}_\nu] = i\hbar \sum_{\kappa} \epsilon_{\mu\nu\kappa} \hat{s}_\kappa . \quad [\hat{\tau}_\mu, \hat{\tau}_\nu] = i\hbar \sum_{\kappa} \epsilon_{\mu\nu\kappa} \hat{\tau}_\kappa .$$

Postulates for a single nucleon $|\psi\rangle$

Postulate 1

A nucleon is completely described by a vector in a Hilbert space.

$$|\psi\rangle \in \mathcal{H}_{\text{space}} \oplus \mathcal{H}_{\text{spin}} \oplus \mathcal{H}_{\text{isospin}} .$$

Q: what is the dimension of this space?

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Postulate 2

The time evolution of the system is governed by the Schrödinger equation

$$i\hbar \frac{d}{dt} |\psi\rangle = \hat{H} |\psi\rangle = \left[\frac{\hat{p}^2}{2m} + V(\vec{r}, \vec{p}, \vec{s}, \vec{\tau}) \right] |\psi\rangle .$$

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Q: what are the stationary states of a completely isolated nucleon?

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Postulate 3

A wavefunction can be interpreted probabilistically.

$$|\langle \vec{r} \sigma \tau | \psi \rangle|^2 = |\psi(\vec{r}, \sigma, \tau)|^2 = P[\text{nucleon at } \vec{r} \text{ with (iso)spin } \sigma(\tau)]$$

By convention: σ and τ are the z-projections of spin and isospin.

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Consider a Hermitian $\hat{O}$ with eigenvalues and eigenstates o_i , $|o_i\rangle$.

Then a measurement of the quantity O will

- always results in an eigenvalue o_i with probability $|\langle O_i | \psi \rangle|^2$.
- collapse the wf to the corresponding eigenstate; $|\psi\rangle \rightarrow |o_i\rangle$

Describing N nucleons

$4N$ measurable quantities = $4N$ Hermitian operators

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Is something missing?

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One-body basis

Consider

$$\mathcal{B}_1 = \{|\alpha\rangle | \alpha = \text{collection of quantum numbers}\}$$

such that it is a set of basis states of $\mathcal{H}_1$.

$$\langle \vec{r}\sigma\tau | \alpha \rangle = \psi_\alpha(\vec{r}, \sigma, \tau) = \text{one-body wavefunction}.$$

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Many-body basis

$$\mathcal{B}_N = \left\{ |1 : \alpha, 2 : \beta, 3 : \gamma, \dots, N : \xi\rangle \middle| \forall \alpha, \beta, \dots \right\}$$

is a basis of $\mathcal{H}_N \equiv \mathcal{H}_1^N = \mathcal{H}_1 \otimes \mathcal{H}_1 \otimes \dots \otimes \mathcal{H}_1$.

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$$\langle 1 : \vec{r}_1\sigma_1\tau_1, 2 : \vec{r}_2\sigma_2\tau_2 \dots | \alpha\beta\gamma \dots \rangle = \psi_\alpha(\vec{r}_1, \sigma_1, \tau_1) \times \psi_\beta(\vec{r}_2, \sigma_2, \tau_2) \times \dots$$

i.e. "particle 1/2/3/... occupies state $\alpha/\beta/\gamma/\dots$ ". Q: Why is $\mathcal{B}_N$ a basis?

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Are these states physically relevant?

Indistinguishable particles

Under the pairwise exchange of particles $i \leftrightarrow j$...

- the accessible states are ...
- the allowed operators are ...

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Q1: Why is only one sign relevant to operators?

Q2: How do $\mathcal{H}_N^{\text{bosons}}$ and $\mathcal{H}_N^{\text{fermions}}$ relate to $\mathcal{H}_N$?

How to construct physically relevant states?

Consider the state

$$|\alpha\beta\gamma\dots\rangle = \frac{1}{\sqrt{N!}} \sum_{p \in P} \pi_p |p(1) : \alpha, p(2) : \beta, p(3) : \gamma, \dots\rangle.$$

Properties

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$N = 4$ example

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Q: How do you define a bosonic $|\alpha\beta\gamma\dots\rangle$?

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Slater determinant

The wavefunction representation of $|\alpha\beta\gamma\dots\rangle$

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The total (spatial, one-body) density

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Could we look at the proton/neutron density separately?

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$$\begin{aligned} \langle \alpha\beta \dots | \rho(\vec{r}) | \alpha\beta \dots \rangle &= \sum_{i=1,N} \langle \alpha\beta \dots | \delta(\vec{r} - \vec{r}_i) | \alpha\beta \dots \rangle \\ &= \frac{1}{N!} \sum_{i=1,N} \sum_{p_1, p_2} \pi_{p_1} \pi_{p_2} \langle p_2(1) : \alpha, \dots | \delta(\vec{r} - \vec{r}_i) | p_1(1) : \alpha, \dots \rangle \\ &= \frac{1}{N!} \sum_{p_1} \sum_{i=1,N} \psi_{p_1(i)}^*(\vec{r}) \psi_{p_1(i)}(\vec{r}) \\ &= \sum_{\mu=\alpha, \beta, \dots} |\psi_{\mu}(\vec{r})|^2. \end{aligned}$$

Only non-zero if $p_2 = p_1$!

We are free to reorder this sum!

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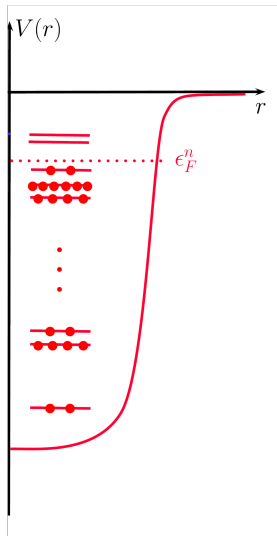
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Pretty complicated, huh!

Interpreting a Slater determinant

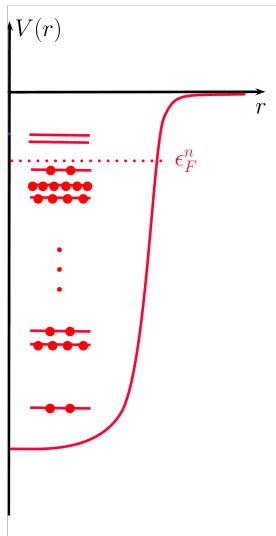
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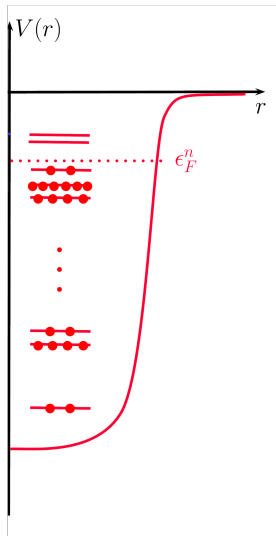
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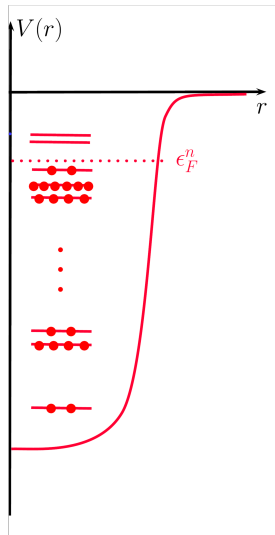
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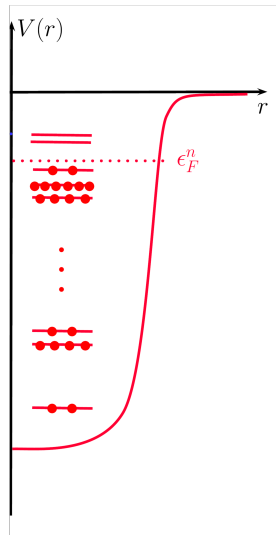
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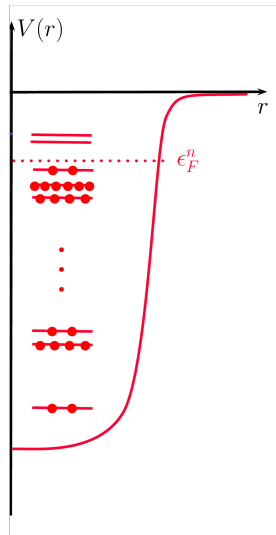
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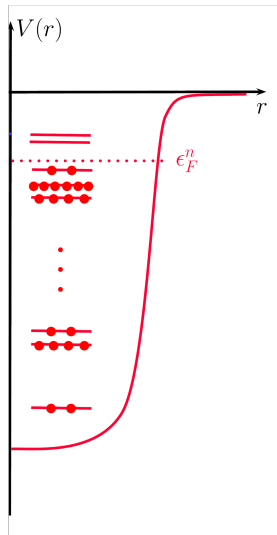
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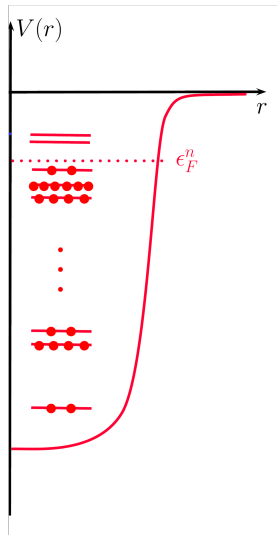
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- But in a Slater determinant these correlations are minimal ...
- A Slater determinant is the simplest wf that is consistent with the Pauli principle.



The example of $N = 2$

$|\alpha\beta\rangle \in \mathcal{H}_2 \rightarrow$ see appendix A in the lecture notes.

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$$\begin{aligned} |\alpha\beta\rangle &\equiv \frac{1}{\sqrt{2!}} \left[|1 : \alpha, 2 : \beta\rangle - |1 : \beta, 2 : \alpha\rangle \right] \\ &= \sqrt{2!} \left(\frac{1}{2!} [1 - P_{12}] \right) |1 : \alpha, 2 : \beta\rangle \\ &= \sqrt{2!} \mathcal{A}_{12} |1 : \alpha, 2 : \beta\rangle. \end{aligned}$$

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Q for next time: can you prove that $|\alpha\beta\rangle$ is normalized if $|\alpha\rangle$ and $|\beta\rangle$ are?

Second quantisation

Fock space

$$\mathcal{F} \equiv \mathcal{H}_0 \oplus \mathcal{H}_1 \oplus \dots \oplus \mathcal{H}_N \oplus \dots$$

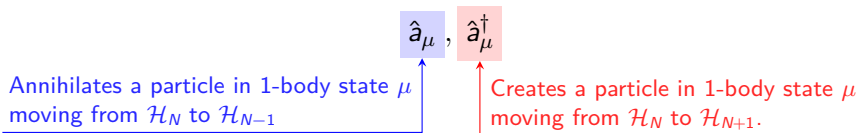
Contains $|0\rangle$

Spanned by basis of Slater determinants $\mathcal{B}_N = \{|\alpha\beta\gamma\dots\rangle\}$

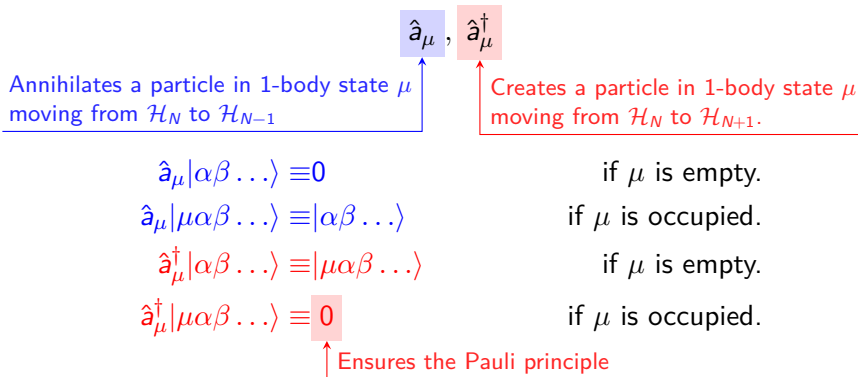
Spanned by single-particle basis $\mathcal{B}_1 = \{|\alpha\rangle\}$

The diagram shows the Fock space $\mathcal{F}$ as a direct sum of subspaces $\mathcal{H}_0, \mathcal{H}_1, \dots, \mathcal{H}_N, \dots$. The subspace $\mathcal{H}_0$ is highlighted in green and has a green arrow pointing to it from the text "Contains $|0\rangle$ ". The subspace $\mathcal{H}_1$ is highlighted in red and has a red arrow pointing to it from the text "Spanned by single-particle basis $\mathcal{B}_1 = \{|\alpha\rangle\}$ ". The subspace $\mathcal{H}_N$ is highlighted in blue and has a blue arrow pointing to it from the text "Spanned by basis of Slater determinants $\mathcal{B}_N = \{|\alpha\beta\gamma\dots\rangle\}$ ".

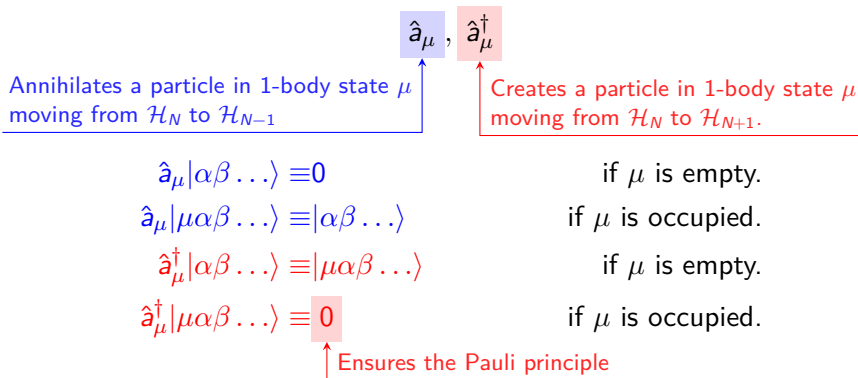
Ladder operators



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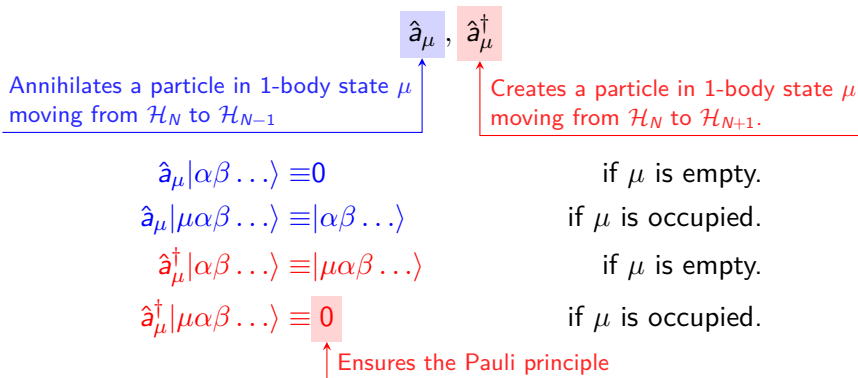
Anticommutation relations

$$\{\hat{a}_\mu^\dagger, \hat{a}_\nu^\dagger\} = 0.$$

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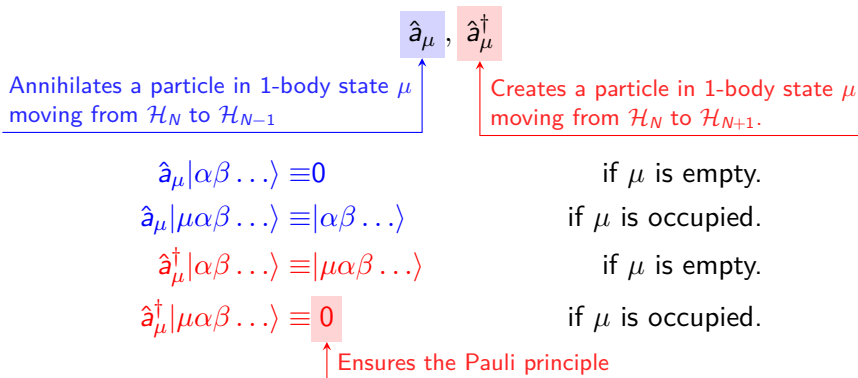


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Q: Does the placement of μ matter in the equations above?

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Q: Does the placement of μ matter in the equations above?

Q for next time: Can you prove the third anticommutation rule?

Slater determinants using ladder operators

Arbitrary Slater determinants in $\mathcal{H}_N$ starting from the vacuum $|0\rangle \in \mathcal{H}_0$

$$|\alpha\beta\gamma\delta\dots\rangle = \hat{a}_\alpha^\dagger \hat{a}_\beta^\dagger \hat{a}_\gamma^\dagger \hat{a}_\delta^\dagger \dots |0\rangle.$$

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Q for next time: Can you prove the antisymmetry of $|\alpha\beta\gamma\delta\dots\rangle$ with ladder operators?

Q for next time: Can you write $\langle\alpha\beta|$ with ladder operators? Can you calculate $\langle\alpha\beta|\gamma\delta\rangle$?

Single-particle basis

All of this depends on our choice for $\mathcal{B}_1 = \{|\mu\rangle\}$, the basis of one-body $\mathcal{H}_1$.
What if we pick a different one, $\mathcal{B}'_1 = \{|\lambda\rangle\}$?

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$$|\mu\rangle = \sum_{\lambda} |\lambda\rangle \langle \lambda | \mu \rangle = \sum_{\lambda} |\lambda\rangle c_{\lambda\mu}.$$

Identity operator in $\mathcal{H}_1$



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Q: what is the transformation rule for annihilation operators?

Q for next time: can you prove C is unitary?

Practical example: position space

Define

$$|\lambda\rangle \equiv |\vec{r}\sigma\tau\rangle \text{ and } \hat{b}_\lambda \equiv \hat{b}_{\vec{r}\sigma\tau}.$$

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plays the role of the transformation C.

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Along similar lines

$$\hat{b}_{\vec{r}\sigma\tau}^\dagger = \sum_\mu \hat{a}_\mu^\dagger \psi_\mu(\vec{r}\sigma\tau) .$$

Choosing $\mathcal{B}_1$

Diagonalise one-body $\hat{h} = \frac{\vec{p}^2}{2m} + V(r)$.
(Iso)spin omitted for simplicity

① Isolated particle

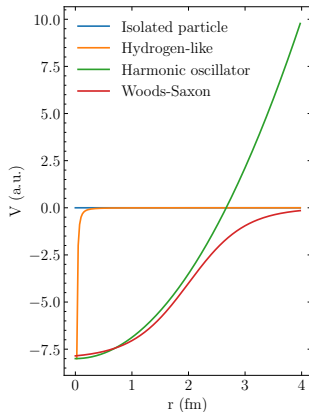
$$\hat{h} = \frac{\vec{p}^2}{2m} \text{ with } \vec{p} = \hbar \vec{k}.$$

$$\psi_{\vec{k}}(\vec{r}) \sim e^{i\vec{k} \cdot \vec{r}}$$

② Hydrogen-like

③ Harmonic oscillator (1D)

④ Woods-Saxon potential



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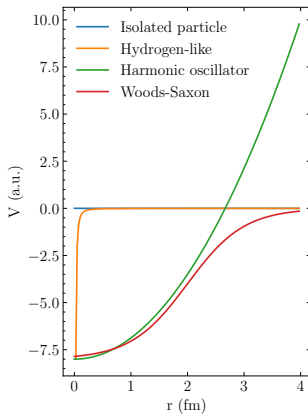
- 1 Isolated particle
- 2 Hydrogen-like

$$\hat{h} = \frac{\vec{p}^2}{2m} - \frac{Ze}{r}$$

$$\psi_{nlm}(r, \theta, \phi) \sim e^{-\rho/2} \rho^\ell L_{n-\ell-1}^{2\ell+1}(\rho) Y_\ell^m(\theta, \phi)$$

with $\rho = \frac{2r}{na_0}$ and $a_0 \equiv$ Bohr radius

- 3 Harmonic oscillator (1D)
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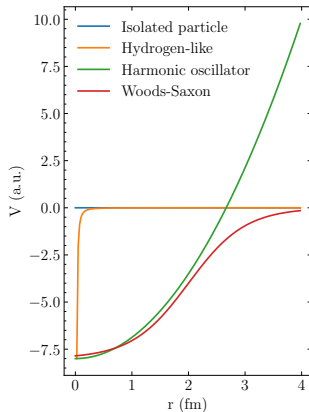
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$$\hat{h} = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2 .$$
$$\psi_n(x) \sim H_n \left(\sqrt{\frac{m\omega}{\hbar}} x \right) e^{\frac{-m\omega}{2\hbar} x^2} .$$

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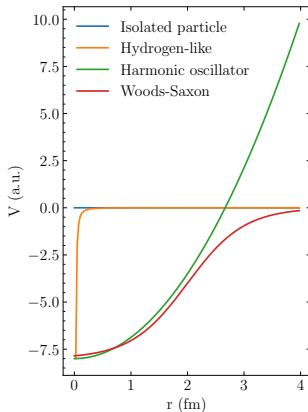
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$$\hat{h} = \frac{\vec{p}^2}{2m} - \frac{V_0}{1 + \exp\left(\frac{r-R_0}{a}\right)} .$$

Complicated, but doable, to solve analytically.



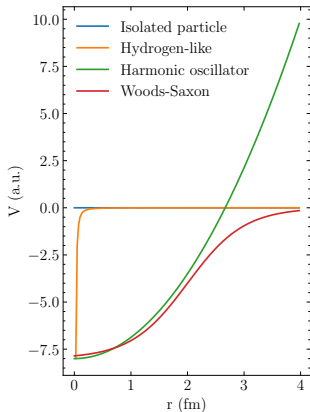
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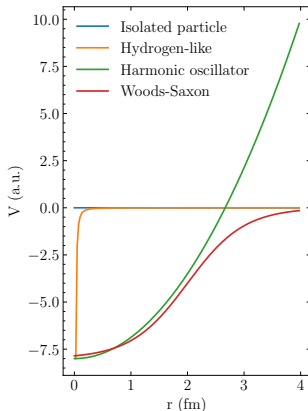
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Q: Which one matches the phenomenology of the strong interaction best?

Q: Does it matter what basis we use for formal manipulations? In practical calculations?

Operators

Operators in first quantisation

$$\text{1-body } F = \sum_{i=1}^N f(i).$$

$$\text{2-body } G = \frac{1}{2} \sum_{i \neq j=1}^N g(i, j).$$

Sums over particles
operators defined on $\mathcal{H}_N$

Operators in **second** quantisation

$$\text{1-body } F = \sum_{\alpha\beta} f_{\alpha\beta} a_{\alpha}^{\dagger} a_{\beta}.$$

$$\text{2-body } G = \frac{1}{2} \sum_{\alpha\beta\gamma\delta} g_{\alpha\beta\gamma\delta} a_{\alpha}^{\dagger} a_{\beta}^{\dagger} a_{\delta} a_{\gamma}.$$

Sums over states in $\mathcal{H}_1$
operators defined on $\mathcal{F}$!

Ordering!

Q for next time: what happens when you act with $\hat{G}$ on states in $\mathcal{H}_0$ and $\mathcal{H}_1$?

Interpretation of operators

Acting with $\hat{F}$ on a state $|\Psi\rangle$

$$|\Phi\rangle = \hat{F}|\Psi\rangle = \sum_{\alpha\beta} f_{\alpha\beta} \hat{a}_{\alpha}^{\dagger} \hat{a}_{\beta} |\Psi\rangle.$$

Diagram annotations:

- Green arrow from $|\Psi\rangle$ to $f_{\alpha\beta}$: Probability amplitude
- Red arrow from $\hat{a}_{\beta}$ to $|\Psi\rangle$: Destroy a particle in β
- Blue arrow from $\hat{a}_{\alpha}^{\dagger}$ to the result: Create a particle in α
- Yellow arrow from $\sum_{\alpha\beta}$ to the result: Sum over all possibilities

What is the probability amplitude? Let's try in $\mathcal{H}_1$

$$\begin{aligned} \langle\mu|\hat{F}|\nu\rangle &= \langle 1 : \mu | \hat{F} | 1 : \nu \rangle = \int \int d\vec{r}_1 d\vec{r}_2 \psi_{\mu}^*(\vec{r}_1) f(\vec{r}_1, \vec{r}_2) \psi_{\nu}(\vec{r}_1) \\ &= \langle\mu| \sum_{\alpha\beta} f_{\alpha\beta} \hat{a}_{\alpha}^{\dagger} \hat{a}_{\beta} |\nu\rangle = f_{\mu\nu} \end{aligned}$$

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Q for next time: Can you express the kinetic energy in first and second quantisation?